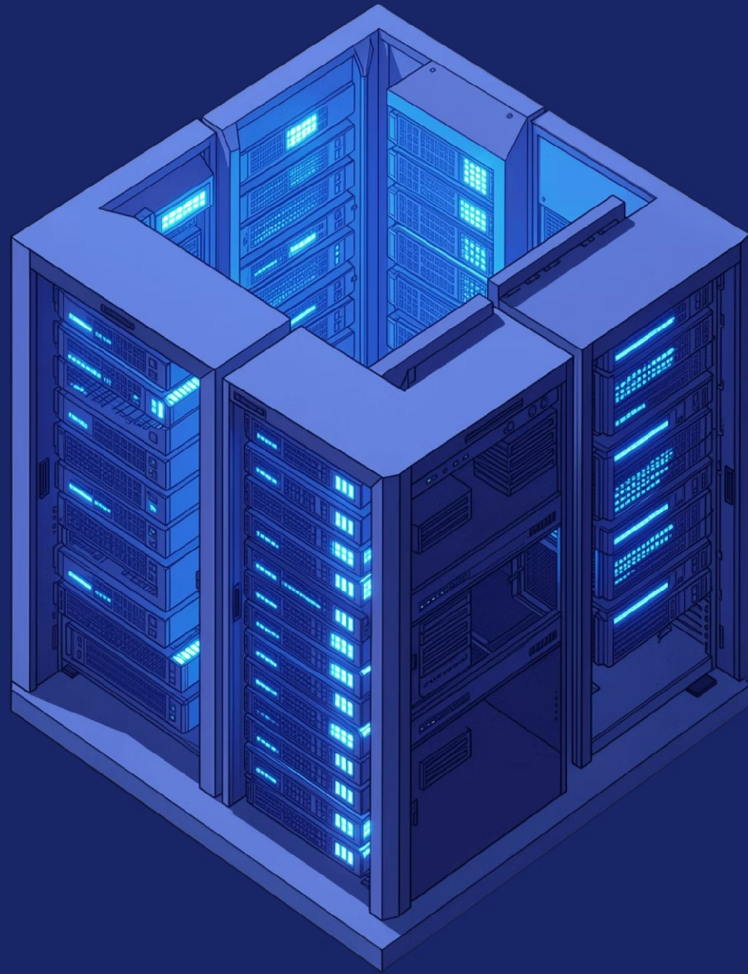




Comparison Between File-Based Systems and Database Management Systems (DBMS)

Using Real-Life Case Studies from Banking and Healthcare

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Introduction

What is a File-Based System?

A file-based system stores data in separate files, where each application manages its own data independently. Since data is stored in multiple locations, maintaining consistency, sharing information, and updating records become difficult.

What is a Database Management System (DBMS)?

A Database Management System (DBMS) is software that stores, organizes, and manages data in a centralized database. It enables secure storage, efficient retrieval, data sharing, and controlled access while minimizing redundancy and improving consistency.

Scattered Storage

File-based systems store information in separate, independently managed files — leading to duplication and inconsistency.

Centralized Storage

DBMS stores information in a unified, centralized database accessible to all authorized users simultaneously.

Modern Standard

DBMS improves data accuracy, security, and efficient management — the standard approach in modern organizations across all industries.

Comparison Between File-Based System and DBMS

Feature	File-Based System	Database Management System (DBMS)
Data Storage	Data stored in separate files managed independently by different applications	Data stored in a centralized database managed by the DBMS
Data Redundancy	High — duplicate copies of data exist in multiple files	Low — duplicate data minimized using normalization techniques
Data Consistency	Difficult to maintain; updates must be made in multiple files	High consistency — all users access the same database
Data Security	Basic security with limited access control	Strong security via authentication, authorization, and encryption
Data Sharing	Sharing between applications is difficult	Easy sharing among multiple users and applications
Backup & Recovery	Manual backup with difficult recovery procedures	Automatic backup and efficient recovery mechanisms
Multi-user Access	Supports limited simultaneous users	Supports multiple concurrent users efficiently
Maintenance	Updating and maintaining files is complex and time-consuming	Easier maintenance through centralized database management

Real-Life Banking Case Study

The Old Way: File-Based Banking

Earlier, banks maintained customer information in separate files at each branch. Updating customer records required changes in multiple files, leading to duplicate data, inconsistencies, and slower banking operations. A customer's balance at one branch would not automatically reflect at another.

Centralized Customer Database

Single source of truth across all branches — no more conflicting records between locations.

Seamless Digital Banking

ATM, internet banking, and mobile banking operate seamlessly through the same centralized database.

The Modern Way: Centralized DBMS

Today, banks use Database Management Systems to maintain a centralized database. All branches access the same customer information in real time, enabling faster transactions and improved customer service across the entire network.

Real-Time Transaction Processing

Instant updates across the entire network — deposits, withdrawals, and transfers reflected immediately.

Security & Fraud Prevention

Role-based access control and fraud prevention mechanisms protect sensitive financial data at every level.

- ❏ **Real-Life Example:** When a customer deposits money in Chennai, the updated balance becomes immediately available at any ATM or bank branch across India — because every branch accesses the same centralized database.

Real-Life Healthcare Case Study

From Paper Records to Electronic Health Records (EHR)

The Old Way: Paper-Based Records

Hospitals previously maintained patient information on paper records or separate files, making it difficult to retrieve medical histories quickly. This often resulted in delayed treatment, repeated diagnostic tests, and duplicate patient records across departments.



Centralized Patient Records

Complete medical history in one secure location — accessible to authorized clinicians instantly.



Secure Medical Data

Encrypted and access-controlled patient information ensures privacy and regulatory compliance.

The Modern Way: EHR & Centralized DBMS

Modern hospitals use Electronic Health Records (EHR) stored in a centralized database. Doctors can instantly access patient history, prescriptions, laboratory reports, and treatment details, improving healthcare quality and efficiency across all departments.



Faster Diagnosis & Treatment

Instant access to lab reports and prescriptions eliminates delays in critical care decisions.



Inter-Department Sharing

Radiology, pharmacy, and wards share data instantly — no manual transfer of files required.



Real-Life Example: If a patient visits another hospital, doctors can instantly access previous medical records, enabling immediate treatment without repeating unnecessary tests.

Advantages of Database Management Systems



Reduced Data Redundancy

Eliminates duplicate information through normalization techniques.



Data Consistency & Integrity

Enforces constraints to maintain accurate, reliable data across all records.



Strong Security

Authentication, authorization, and encryption protect sensitive organizational data.



Multi-User Concurrency

Supports simultaneous access by multiple users without conflict or data corruption.



Automatic Backup & Recovery

Built-in disaster recovery ensures business continuity even after system failures.



Fast Query Retrieval

Structured Query Language (SQL) enables rapid, precise data access at scale.



Simplified Administration

Centralized management reduces maintenance complexity across the organization.



Scalability

Easily accommodates growing volumes of organizational data without restructuring.

Conclusion

File-based systems are suitable only for small-scale applications but become inefficient as the volume of data increases.

Database Management Systems overcome these limitations by providing centralized storage, better security, reduced redundancy, faster data retrieval, and reliable backup mechanisms.

Today, industries such as banking and healthcare rely heavily on DBMS to ensure efficient operations, accurate information, and improved customer service.

Key Takeaways

Centralized Data Management

One unified system replaces scattered file stores — eliminating silos and inconsistencies across the organization.

Better Security and Privacy

Robust access control and encryption protect sensitive information from unauthorized access and breaches.

Faster Access to Information

Efficient SQL queries deliver results in real time — enabling faster decisions and improved service delivery.

Reduced Redundancy & Improved Consistency

Normalization ensures data integrity — every record is accurate, up-to-date, and free of duplication.

Essential Technology for Modern Organizations

DBMS is the backbone of banking, healthcare, and beyond — indispensable for any data-driven enterprise.



Thank You

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DBMS Assessment – College
Presentation